License Plate Detection via Transfer Learning with YOLOv5

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Abstract

We study automatic license plate localization in road-scene images. The task is straightforward to state: given an image, detect whether a license plate is present, and if so, draw a bounding box around it. We achieve this by fine-tuning YOLOv5s [5], pretrained on MS-COCO, into a single-class plate detector. Training used the public Kaggle license-plate dataset (approximately 27 000 annotated images) at 640×640 resolution for 20 epochs. Because YOLOv5s already provides strong defaults, minimal hyper-parameter tuning was needed beyond selecting the Adam optimizer and keeping the standard augmentation schedule. Qualitative evaluation on held-out images shows confident, tightly localized predictions (conf ≥ 0.9) on well-lit plates, with remaining false positives easily suppressed by a modest confidence threshold increase. The project demonstrates that off-the-shelf transfer learning on a general-purpose detector is sufficient for a practical plate-detection pipeline.

1. Motivation

We were interested in building an automatic license plate detector motivated by a concrete goal: running the model on Lucian's dashcam footage to see how well a commodity detector handles real, unscripted highway driving. License plate detection is the first stage of any Automatic License Plate Recognition (ALPR) pipeline — every downstream step (character segmentation, OCR, database lookup) depends on getting the bounding box right. We chose to tackle this stage first before extending to full recognition.

2. Related Work

YOLO family. The YOLO [12] line frames detection as a single dense-grid regression. YOLOv2–v3 [10, 11] added anchor boxes and multi-scale heads; YOLOv4 [2] added Mosaic augmentation and CIoU loss. YOLOv5 [5], our choice, combines a CSPDarknet backbone with a PANet neck [8] and three detection heads at strides 8, 16, and 32.

Alternative architectures. SSD [9] and RetinaNet [7] are competitive one-stage detectors. Faster R-CNN [13] and DETR [3] offer stronger accuracy baselines at higher inference cost. EfficientDet [17] jointly scales backbone and neck for a range of compute budgets.

Plate-specific methods. WPOD-NET [15] predicts a per-detection affine warp to rectify each plate into a canonical view. Our work deliberately uses a general-purpose detector to measure how well commodity transfer learning performs without plate-specific inductive bias.

3. Technical Approach

3.1. Dataset

We use the Kaggle `fareselmenshawii/license-plate-dataset`, which ships in the standard YOLO layout: an `images/{train,val}` directory tree paired with a `labels/{train,val}` tree of per-image text files. Each label file contains one row per plate with the YOLO-normalized tuple (c, x, y, w, h), where $c = 0$ is the single plate class, (x, y) is the box center, and (w, h) is the box size, all expressed as fractions of image width and height. The dataset is downloaded at runtime with the `kagglehub` client.

Each image is loaded with Pillow, resized to 640×640, converted to a float tensor in $[0, 1]$, and permuted from HWC to CHW before being stacked into a mini-batch. Labels are concatenated across the batch with a leading image-index column into the six-column (i, c, x, y, w, h) layout that YOLOv5's loss expects.

3.2. Model

YOLOv5s consists of a CSPDarknet backbone, a PANet neck, and three detection heads. The backbone interleaves convolutional blocks with cross-stage partial (CSP) connections to reduce redundant gradient paths. The neck propagates features top-down and bottom-up so that each head sees both semantically rich and spatially precise representations. The three heads produce anchor-based predictions at strides 8, 16, and 32; the stride-8 head does most of the work for small plates.

We instantiate the model from `models/yolov5s.yaml` with `nc=1` and load the official `yolov5s.pt` COCO checkpoint. Because the final $1{\times}1$ convolutions of the three heads have incompatible shapes between the 80-class source and the 1-class target, we filter the pretrained state dict to retain only keys whose shape matches the new model. This means the backbone and neck are warm-started from COCO features while the detection heads train from scratch on plates — a simple and effective form of partial transfer learning.

3.3. Loss

We use YOLOv5's `ComputeLoss`:

$$\mathcal{L} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{obj}}\mathcal{L}_{\text{obj}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}}$$

The box term is CIoU [19], which augments plain IoU with a distance penalty between box centers and an aspect-ratio consistency term. The objectness and class terms are binary cross-entropy. In the single-class setting the class term is effectively inert, so training signal flows almost entirely through the objectness and box-regression channels.

3.4. Training

Training ran for 20 epochs with batch size 32 using the Adam optimizer at a constant learning rate of 1×10^{-3} and 640×640 input resolution. Because YOLOv5s provides strong default hyperparameters, minimal tuning was required beyond these basic choices.

The standard YOLOv5 augmentation schedule is kept: mosaic stitching (four images per sample), horizontal flip with probability 0.5, HSV jitter (hue 0.015, saturation 0.7, value 0.4), random scale ± 0.5, and random translation ± 0.1. Vertical flip, shear, rotation, and perspective warping are disabled to respect the canonical up-orientation of license plates.

3.5. Inference

At test time we apply non-maximum suppression with confidence threshold 0.1 and IoU threshold 0.45. The deliberately low confidence threshold exposes the raw detector output including its weakest candidate boxes. Each surviving detection is drawn as a bounding box annotated with its confidence score. No additional post-processing is applied — no aspect-ratio filter, no minimum-area filter — so we can measure what the bare detector has learned.

4. Results

True positives. Figure 1a shows a marked police sedan whose rear plate is cleanly boxed at 0.92 confidence under even, shadowless lighting. Figure 1b shows a pickup truck photographed in direct sunlight; the detector still fires at 0.91 despite the bright glare washing out detail around the plate. Figure 1c is the most demanding case: a small, shadowed plate positioned near the image boundary. The stride-8 detection head, responsible for small objects, recovers it at 0.90.

False positives. Figure 1d shows a spurious detection on a hanging commercial sign: a compact, text-bearing rectangle, exactly the kind of texture the model is trained to fire on. Figure 1e fires on a windowed building facade, another high-contrast rectangular grid. Both are intuitive failure modes, and crucially both have confidences well below any true detection (0.37 and 0.14 respectively). A confidence threshold of 0.5 removes both while leaving all three correct localizations intact.

4.1. Dashcam Video Evaluation

We are currently running the retrained detector on frame-by-frame extracts from Lucian's dashcam footage. This test covers continuous, unscripted highway driving and exposes conditions absent from the static training set: motion blur, partial occlusion, rapid distance changes, and lens glare. Sample frames, per-condition detection rates, and temporal consistency analysis will be included in the next revision of this report.

4.2. Conclusion

Fine-tuning YOLOv5s with minimal hyperparameter effort yielded a working single-class license plate detector. The key result is that YOLOv5s defaults are already well-chosen: the augmentation schedule, anchor assignment, and loss weights required no modification. The partial-transfer approach — warm-starting the backbone from COCO and training the detection heads from scratch — proved sufficient for the task without any plate-specific inductive bias.

The main remaining failure mode is low-confidence false positives on rectangular signage, which a simple confidence-threshold increase eliminates in practice.

4.3. Future Work

The natural next step is to extend the pipeline from detection to full recognition: given a tightly cropped plate bounding box, apply OCR to extract the alphanumeric plate number. This would complete an end-to-end ALPR pipeline and allow evaluation against ground-truth plate strings from the dashcam footage.

(a) Police sedan, conf $= 0.92$. Tight box on a standard rear plate under even lighting.

(b) Pickup in bright sunlight, conf $= 0.91$. Clean detection despite direct glare.

(c) Small shadowed plate near image boundary, conf $= 0.90$. The stride-8 head recovers the plate.

(d) False positive on a hanging commercial sign, conf $= 0.37$. Suppressed at threshold 0.5.

(e) False positive on a rectangular building facade, conf $= 0.14$. Suppressed at threshold 0.5.

Figure 1. Detector output on five held-out validation images at conf ≥ 0.1. **Top row**: all three true plates are recovered at high confidence (≥ 0.90) with tight bounding boxes across varied conditions. **Bottom row**: both false positives fire on high-contrast rectangular objects (signage, building windows) but at low confidence (0.14–0.37); a deployment threshold of 0.5 removes them while preserving every correct detection.

Figure 2. Dashcam video results — to be included in next revision.

References

[1] AMD/Xilinx. EfficientDet_D2 – Xilinx AI SDK. https://docs.amd.com/r/en-US/ug1354-xilinx-ai-sdk/EfficientDet_D2, 2023.

[2] Alexey Bochkovskiy, Chien-Yao Wang, and Hong-Yuan Mark Liao. Yolov4: Optimal speed and accuracy of object detection. *arXiv preprint arXiv:2004.10934*, 2020.

[3] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European Conference on Computer Vision (ECCV)*, pages 213–229, 2020.

[4] Md Fahim Bin Amin. 100k vehicle dashcam image dataset. Kaggle, https://www.kaggle.com/datasets/mdfahimbinamin/100k-vehicle-dashcam-image-dataset, 2023.

[5] Glenn Jocher. YOLOv5 by Ultralytics. https://github.com/ultralytics/yolov5, 2020.

[6] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2117–2125, 2017.

[7] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.

[8] Shu Liu, Lu Qi, Haifang Qin, Jianping Shi, and Jiaya Jia. Path aggregation network for instance segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8759–8768, 2018.

[9] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C. Berg. SSD: Single shot multibox detector. In *European Conference on Computer Vision (ECCV)*, pages 21–37, 2016.

[10] Joseph Redmon and Ali Farhadi. YOLO9000: Better, faster, stronger. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7263–7271, 2017.

[11] Joseph Redmon and Ali Farhadi. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*, 2018.

[12] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 779–788, 2016.

[13] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

[14] Roboflow. Autodistill: Use large foundation models to train small, specialized models. https://github.com/autodistill/autodistill, 2023.

[15] Sérgio Montazzolli Silva and Cláudio Rosito Jung. License plate detection and recognition in unconstrained scenarios. In *European Conference on Computer Vision (ECCV)*, pages 580–596, 2018.

[16] Solesensei. BDD100K dataset. Kaggle, https://www.kaggle.com/datasets/solesensei/solesensei_bdd100k, 2020.

[17] Mingxing Tan, Ruoming Pang, and Quoc V. Le. Efficientdet: Scalable and efficient object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10781–10790, 2020.

[18] Fisher Yu, Haofeng Chen, Xin Wang, Wenqi Xian, Yingying Chen, Fangchen Liu, Vashisht Madhavan, and Trevor Darrell. BDD100K: A diverse driving dataset for heterogeneous multitask learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2636–2645, 2020. http://bdd-data.berkeley.edu/.

[19] Zhaohui Zheng, Ping Wang, Wei Liu, Jinze Li, Rongguang Ye, and Dongwei Ren. Distance-IoU loss: Faster and better learning for bounding box regression. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020.